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VEHICLE CARGO SECURING SYSTEM

Abstract

A vehicle cargo securing system includes a plurality of stabilizers each having a stand and a base. The stands are pivotably coupled to the bases. The stands are pivotable back-and-forth between a stowed position where the stand is disposed within a groove of a cargo bed floor and a cargo support position where the stand is pivoted to a position where the stand projects out of the groove and away from the cargo bed floor.

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Background/Summary

TECHNICAL FIELD

[0001] This disclosure relates generally to a securing system that is used to secure cargo such as,

for example, bicycles and motorcycles.

BACKGROUND

[0002] Vehicles can include many areas utilized to transport cargo. Pickup trucks, for example, include a cargo bed. Systems can be utilized to secure cargo within the cargo bed.

SUMMARY

[0003] In some aspects, the techniques described herein relate to a vehicle cargo securing system, including: a plurality of stabilizers each having a stand and a base, the stand pivotably coupled to the base, the stand pivotable back-and-forth between a stowed position where the stand is disposed within a groove of a cargo bed floor and a cargo support position where the stand is pivoted to a position where the stand projects out of the groove and away from the cargo bed floor.

[0004] In some aspects, the techniques described herein relate to a vehicle cargo securing system, wherein the stand and the cargo bed floor are disposed along planes that are aligned with each other when the stand is in the stowed position, where the stand and the cargo bed floor are disposed along planes that are angled relative to each other when the stand is in the cargo support position.

[0005] In some aspects, the techniques described herein relate to a vehicle cargo securing system, wherein each of the stabilizers within the plurality of stabilizers includes a hinge pivotably coupling the stand to the base.

[0006] In some aspects, the techniques described herein relate to a vehicle cargo securing system, wherein the stand is lockable in the cargo support position.

[0007] In some aspects, the techniques described herein relate to a vehicle cargo securing system, wherein the base is secured to the cargo bed floor with at least one mechanical fastener.

[0008] In some aspects, the techniques described herein relate to a vehicle cargo securing system, wherein each of the stabilizers within the plurality of stabilizers includes at least one magnet that holds the stand in the stowed position.

[0009] In some aspects, the techniques described herein relate to a vehicle cargo securing system, wherein the plurality of stabilizers includes a first stabilizer and a second stabilizer that are configured to support a wheel of a vehicle that is between the first stabilizer and second stabilizer when both the stand of the first stabilizer is in the cargo support position and the stand of the second stabilizer is in the cargo support position.

[0010] In some aspects, the techniques described herein relate to a vehicle cargo securing system, further including a strap coupled to the stand of the first stabilizer and to the stand of the second stabilizer when both the stand of the first stabilizer is in the cargo support position and the stand of the second stabilizer is in the cargo support position, the strap extending between a plurality of spokes of the wheel of the vehicle.

[0011] In some aspects, the techniques described herein relate to a vehicle cargo securing system, wherein the vehicle is a bicycle and the wheel is a front wheel of the bicycle.

[0012] In some aspects, the techniques described herein relate to a vehicle cargo securing system, wherein the plurality of stabilizers are a plurality of floor-mounted stabilizers that are secured to the cargo bed floor adjacent to a cargo bed wall, and further including at least one wall-mounted stabilizer secured to an upper region of the cargo bed wall, the plurality of floor-mounted stabilizers configured to interface with a lower portion of the wheel, the at least one wall-mounted stabilizer configured to interface with an upper portion of the wheel.

[0013] In some aspects, the techniques described herein relate to a vehicle cargo securing system, wherein the at least one wall-mounted stabilizer includes a mounting bracket and a stabilizing bracket that is pivotably coupled to the mounting bracket, the at least one wall-mounted stabilizer further including a strap that is anchored to the mounting bracket and is configured to extend through a plurality of spokes of the wheel of the vehicle.

[0014] In some aspects, the techniques described herein relate to a vehicle cargo securing system, wherein the cargo bed wall is a front wall of the cargo bed.

[0015] In some aspects, the techniques described herein relate to a vehicle cargo securing method,

including: pivoting a first stand of a first stabilizer from a stowed position to a cargo support position, the first stand received within a first groove of a cargo bed floor when the first stand is in the stowed position, the first stand in the cargo support position pivoted to project out of the first groove and away from the cargo bed floor; pivoting a second stand of a second stabilizer from a stowed position to a cargo support position, the second stand received within a second groove of a cargo bed floor when the second stand is in the stowed position, the second stand in the cargo support position pivoted to project out of the second groove and away from the cargo bed floor; and supporting a wheel of a vehicle between the first stand of the first stabilizer in the cargo support position and the second stand of the second stabilizer in the cargo support position.

[0016] In some aspects, the techniques described herein relate to a vehicle cargo securing method, wherein the wheel is a front wheel of a two-wheeled vehicle.

[0017] In some aspects, the techniques described herein relate to a vehicle cargo securing method, wherein the wheel is a front wheel of a bicycle.

[0018] In some aspects, the techniques described herein relate to a vehicle cargo securing method, further including supporting the wheel using a strap that is connected to the first stand and the second stand and extends through a plurality of spokes of the wheel.

[0019] In some aspects, the techniques described herein relate to a vehicle cargo securing method, further including securing a first base of the first stabilizer to the cargo bed floor adjacent to a cargo bed wall, and securing a second base of the second stabilizer to the cargo bed floor adjacent to the cargo bed wall.

[0020] In some aspects, the techniques described herein relate to a vehicle cargo securing method, further including supporting an upper portion of the wheel using at least one wall-mounted stabilizer secured to an upper region of the cargo bed wall.

[0021] In some aspects, the techniques described herein relate to a vehicle cargo securing method, further including supporting the upper region using a strap of the at least one wall-mounted stabilizer, the strap extending through spokes of the wheel of the vehicle.

[0022] The embodiments, examples and alternatives of the preceding paragraphs, the claims, or the following description and drawings, including any of their various aspects or respective individual features, may be taken independently or in any combination. Features described in connection with one embodiment are applicable to all embodiments, unless such features are incompatible.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0023] The various features and advantages of the disclosed examples will become apparent to those skilled in the art from the detailed description. The figures that accompany the detailed description can be briefly described as follows:

[0024] FIG. 1 illustrates a perspective view of a vehicle having a cargo bed and carrying cargo that is secured using a cargo securing system according to an exemplary aspect of the present disclosure.

[0025] FIG. 2 shows a perspective view of the cargo securing system shown in FIG. 1 with the cargo securing system having stands of stabilizers in a stowed position.

[0026] FIG. 3 illustrates a section view at line 3-3 in FIG. 2 showing the stands in the stowed position and a cargo support position of the stands shown in broken lines.

[0027] FIG. 4 illustrates the perspective view of the cargo securing system of FIG. 2 having the stands of the stabilizers in a cargo support position.

[0028] FIG. 5 illustrates the perspective view of the cargo securing system of FIG. 4 having the stands of the stabilizers in the cargo support position and with cargo.

[0029] FIG. 6 illustrates a close-up perspective view of the cargo securing system of FIG. 5.

[0030] FIG. 7 illustrates a top view of the cargo securing system in the position of FIG. 5 along with other stabilizers and cargo.

DETAILED DESCRIPTION

[0031] This disclosure details exemplary embodiments of a vehicle cargo securing system. The embodiments utilize the securing systems to secure a bicycle. In other examples, the securing system secures other types of cargo.

[0032] With reference to FIGS. 1-6, a vehicle **10** includes a cargo bed **14** aft of a passenger compartment **18**. The cargo bed **14** includes a front wall **22**, opposing sidewalls **26**, and a tailgate **30**. The cargo bed **14** includes a cargo bed floor **34** having a plurality of grooves **38** or corrugations. The grooves **38** can enhance structural integrity of the floor **34** and the overall cargo bed **14**.

[0033] A vehicle cargo securing system **50** is used to secure cargo within the cargo bed **14**. That cargo securing system **50** can provide tie-down locations for securing cargo.

[0034] In this exemplary embodiment of this disclosure, the vehicle cargo securing system **50** includes a plurality of stabilizers **54** mounted to the cargo bed floor **34**. In this particular embodiment, the securing system **50** utilizes two stabilizers **54**.

[0035] Each of the stabilizers **54** includes a base **58** that is pivotably connected to a stand **62**. The base **58** and the stand **62** can be a rigid polymer-based material.

[0036] The stabilizers **54** additionally include a hinge **66** that pivotably connects together the base **58** and the stand **62**. The stand **62** of each of the stabilizers **54** is pivotable relative to the respective base **58** back-and-forth between a stowed position shown in FIGS. 2 and 3 and a cargo support position shown in FIGS. 1 and 4-6.

[0037] In the stowed position, the stands **62** are each disposed within one of the grooves **38** of the cargo bed floor **34**. In the stowed position, the example stands **62** along with the remaining portions of the stabilizers **54** are fully received within the grooves **38**. That is, the example stabilizers **54** do not protrude above the cargo bed floor **34** when the stands **62** are in the stowed position.

[0038] When the stands **62** are in the stowed position, the stands **62** are disposed along a plane P.sub.1 that is aligned with a plane P.sub.2 of the cargo bed floor **34**. As the stand **62** is disposed within the groove **38** when in the stowed position, the stand **62** does not interfere with cargo when the stabilizers **54** are not needed to secure cargo. That is, a user can load and unload sheets of drywall or plywood from the cargo bed **14** without the stand **62** blocking the sheets from sliding along the cargo bed floor **34**.

[0039] In the exemplary embodiment, the stands **62** include magnets **70** that help to hold the stands **62** in the stowed position due to the magnetic attraction to the cargo bed floor **34**. The magnets **70** can be encapsulated to inhibit corrosion.

[0040] In some examples, bumpers are positioned between the stands **62** and the cargo bed floor **34** to dampen movement of the stands **62** back to the stowed position.

[0041] To utilize the securing system **50** to secure cargo, the stands **62** are pivoted from the stowed position to the cargo support position shown. The hinge **66** of the stabilizers **54** could be a lockable hinge or a controlled torque hinge so that the stand **62** can maintain itself in the cargo support position.

[0042] When in the cargo support position, the plane P.sub.1 is angled relative to the plane P.sub.2 of the cargo bed floor **34**. When the stabilizers **54** are in the cargo support position, the stands **62** project out of the respective grooves **38** away from the cargo bed floor **34**. A wheel **74** of cargo **78** can be positioned between the stands **62** of the stabilizers **54** to secure the cargo **78**. The wheel **74** is stabilized by the stabilizers **54**.

[0043] In some examples, a strap **82** can connect to the stands **62** of each of the stabilizers **54**. The strap **82** can extend through spokes **86** of the front wheel **74**. In this example, the cargo **78** is a two-wheeled vehicle, such as a bicycle. The front wheel **74** of the cargo **78** has a plurality of bicycle spokes.

[0044] In the exemplary embodiment, the stabilizers **54** are anchored to the cargo bed floor **34** using a plurality of mechanical fasteners **90** that extend through the bases **58** to engage the cargo bed floor **34**. As the stabilizers **54** are anchored to the cargo bed floor **34**, the stabilizers **54** can be considered floor-mounted stabilizers. The stabilizers **54** can be repositioned at various areas of the cargo bed floor **34** as required.

[0045] The exemplary securing system **50** includes at least one wall-mounted stabilizer **94** in addition to the stabilizers **54**. In the exemplary embodiment, the wall-mounted stabilizer **94** includes a mounting bracket **98** and a stabilizing bracket **102** that is pivotably coupled to the mounting bracket **98**. The wall-mounted stabilizer **94** can additionally include a strap **106** that can loop through spokes **86** of the wheel **74** of the cargo **78**.

[0046] The wall-mounted stabilizer **94** is configured to interface with an upper portion of the front wheel **74** while the stabilizers **54** are configured to interface with a lower portion of the wheel **74**. Upper and lower, for purposes of this disclosure, are with reference to ground in a general orientation of the vehicle **10** during operation when carrying the cargo **78**.

[0047] The mounting bracket **98** of the wall-mounted stabilizer **94** can be pivotably coupled to the stabilizing bracket **102**. This enables the stabilizing bracket **102** to be adjusted to accommodate different sizes and position for the wheels **74**.

[0048] The wall-mounted stabilizer **94** is coupled directly to the front wall **22** of the cargo bed **14**, and in particular, to an upper region—here an upper edge—of the front wall **22**. The stabilizers **54** are secured to the cargo bed floor **34** at a position adjacent a bottom of the front wall **22**.

[0049] Although described in connection with the front wall **22**, the securing system **50** of this disclosure could be utilized in connection with other cargo bed walls, such as the sidewalls **26** or even the tailgate **30**.

[0050] Further, as shown in FIG. 7, a plurality of the cargo securing system **50** could be used to secure respective cargo within the cargo bed **14** at various cross-vehicle positions.

[0051] The preceding description is exemplary rather than limiting in nature. Variations and modifications to the disclosed examples may become apparent to those skilled in the art that do not necessarily depart from the essence of this disclosure. Thus, the scope of protection given to this disclosure can only be determined by studying the following claims.

Claims

1. A vehicle cargo securing system, comprising: a plurality of stabilizers each having a stand and a base, the stand pivotably coupled to the base, the stand pivotable back-and-forth between a stowed position where the stand is disposed within a groove of a cargo bed floor and a cargo support position where the stand is pivoted to a position where the stand projects out of the groove and away from the cargo bed floor.
2. The vehicle cargo securing system of claim 1, wherein the stand and the cargo bed floor are disposed along planes that are aligned with each other when the stand is in the stowed position, where the stand and the cargo bed floor are disposed along planes that are angled relative to each other when the stand is in the cargo support position.
3. The vehicle cargo securing system of claim 1, wherein each of the stabilizers within the plurality of stabilizers includes a hinge pivotably coupling the stand to the base.
4. The vehicle cargo securing system of claim 1, wherein the stand is lockable in the cargo support position.
5. The vehicle cargo securing system of claim 1, wherein the base is secured to the cargo bed floor with at least one mechanical fastener.
6. The vehicle cargo securing system of claim 1, wherein each of the stabilizers within the plurality of stabilizers includes at least one magnet that holds the stand in the stowed position.
7. The vehicle cargo securing system of claim 1, wherein the plurality of stabilizers includes a first

stabilizer and a second stabilizer that are configured to support a wheel of a vehicle that is between the first stabilizer and second stabilizer when both the stand of the first stabilizer is in the cargo support position and the stand of the second stabilizer is in the cargo support position.

8. The vehicle cargo securing system of claim 7, further comprising a strap coupled to the stand of the first stabilizer and to the stand of the second stabilizer when both the stand of the first stabilizer is in the cargo support position and the stand of the second stabilizer is in the cargo support position, the strap extending between a plurality of spokes of the wheel of the vehicle.

9. The vehicle cargo securing system of claim 8, wherein the vehicle is a bicycle and the wheel is a front wheel of the bicycle.

10. The vehicle cargo securing system of claim 7, wherein the plurality of stabilizers are a plurality of floor-mounted stabilizers that are secured to the cargo bed floor adjacent to a cargo bed wall, and further comprising at least one wall-mounted stabilizer secured to an upper region of the cargo bed wall, the plurality of floor-mounted stabilizers configured to interface with a lower portion of the wheel, the at least one wall-mounted stabilizer configured to interface with an upper portion of the wheel.

11. The vehicle cargo securing system of claim 10, wherein the at least one wall-mounted stabilizer includes a mounting bracket and a stabilizing bracket that is pivotably coupled to the mounting bracket, the at least one wall-mounted stabilizer further comprising a strap that is anchored to the mounting bracket and is configured to extend through a plurality of spokes of the wheel of the vehicle.

12. The vehicle cargo securing system of claim 10, wherein the cargo bed wall is a front wall of a cargo bed.

13. A vehicle cargo securing method, comprising: pivoting a first stand of a first stabilizer from a stowed position to a cargo support position, the first stand received within a first groove of a cargo bed floor when the first stand is in the stowed position, the first stand in the cargo support position pivoted to project out of the first groove and away from the cargo bed floor; pivoting a second stand of a second stabilizer from a stowed position to a cargo support position, the second stand received within a second groove of a cargo bed floor when the second stand is in the stowed position, the second stand in the cargo support position pivoted to project out of the second groove and away from the cargo bed floor; and supporting a wheel of a vehicle between the first stand of the first stabilizer in the cargo support position and the second stand of the second stabilizer in the cargo support position.

14. The vehicle cargo securing method of claim 13, wherein the wheel is a front wheel of a two-wheeled vehicle.

15. The vehicle cargo securing method of claim 13, wherein the wheel is a front wheel of a bicycle.

16. The vehicle cargo securing method of claim 13, further comprising supporting the wheel using a strap that is connected to the first stand and the second stand and extends through a plurality of spokes of the wheel.

17. The vehicle cargo securing method of claim 13, further comprising securing a first base of the first stabilizer to the cargo bed floor adjacent to a cargo bed wall, and securing a second base of the second stabilizer to the cargo bed floor adjacent to the cargo bed wall.

18. The vehicle cargo securing method of claim 17, further comprising supporting an upper portion of the wheel using at least one wall-mounted stabilizer secured to an upper region of the cargo bed wall.

19. The vehicle cargo securing method of claim 18, further comprising supporting the upper region using a strap of the at least one wall-mounted stabilizer, the strap extending through spokes of the wheel of the vehicle.
